HOSPITAL MANAGEMENT SYSTEM

By

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1. Introduction

1.1 General Information

Project Name: Hospital Management System **Starting Date:** February 2025

Controlling Agency: **Final Date:** July 2025

Prepared by: Ali Hamza, Iqra, Attiqa **Authorized by:**

1.2 Purpose

Introduction*:*

This proposal outlines the development of a comprehensive **Hospital Management System** (HMS) aimed at optimizing the administrative, financial, and clinical operations within a hospital or healthcare facility. The system is designed to serve as an integrated platform that centralizes data and automates routine processes, thereby improving efficiency, accuracy, and decision-making across departments. By digitizing key workflows—such as patient registration, appointment scheduling, billing, inventory management, and medical recordkeeping—the HMS will significantly reduce the reliance on manual paperwork and minimize the risk of human error.

In addition to enhancing operational efficiency, the HMS will contribute to improved patient care by providing healthcare professionals with instant access to accurate and up-to-date patient information, diagnostic reports, treatment histories, and prescriptions. The system will also include role-based access controls to ensure data privacy and security, complying with healthcare standards and regulations. Furthermore, the HMS will offer analytical tools for reporting and performance monitoring, enabling hospital administrators to make informed strategic decisions.

1.3 Document Conventions

Conventions for main title:

- Font face: Time New Roman (Body)
- Font Style: Bold Italic
- Font Size: 16

Conventions for Sub title:

- Font face: Time New Roman (Body)
- Font Style: Bold Italic
- Font Size: 14

Conventions for Body:

- Font face: Time New Roman (Body)
- Font Style: Bold Italic
- Font Size: 12

1.4 Intended Audience:

This system is intended for a wide range of users within a healthcare setup. Each type of user interacts with the system in a unique way based on their role and responsibilities.

Audiences Include:

1) **Hospital Administration:**
   a) To monitor hospital performance and manage resources.
   b) To generate analytical reports and ensure policy compliance.
2) **Doctors:**
   a) To view and update patient records.
   b) To manage appointments and prescriptions.
3) **Nursing Staff:**
   a) To assist in patient care and maintain treatment records.
   b) To check schedules and medication instructions.
4) **Receptionists / Front Desk Staff:**
   a) To register new patients and manage their appointments.
   b) To respond to inquiries and manage check-ins/check-outs.
5) **Pharmacy Staff:**
   a) To manage medicine stock, prescriptions, and issuing of drugs.
6) **Laboratory Technicians:**
   a) To update test results and manage test schedules.
7) **IT Department:**
   a) For system maintenance, backup, and data security.
8) **Health Authorities / Government Agencies:**
   a) For monitoring public health trends and compliance reporting.

1.5 Literature Review

The development of Hospital Management Systems (HMS) has been widely studied and implemented in various healthcare settings to improve efficiency, data management, and patient care. Existing research highlights the transition from traditional paper-based systems to digital platforms, emphasizing the benefits of automation in reducing errors, saving time, and improving accessibility of patient records.Several existing studies and software implementations emphasize the importance of technology in improving hospital workflows and reducing human error. Traditional paper-based systems are prone to data loss, duplication, and inefficiencies. Recent research highlights benefits of digital systems in:

i. Reducing patient wait times
ii. Enhancing data accuracy
iii. Enabling seamless communication among departments
iv. Supporting decision-making with data analytics
v. Systems such as **OpenMRS**, **Meditech**, and **Epic** have been successfully implemented worldwide but often face issues related to high cost, complexity, and customization for smaller hospitals. This project seeks to build a cost-effective, scalable solution tailored for mid-sized hospitals, learning from both successful and failed implementations.

2 Problem Statement

Following are the issues and deficiencies in previous **Hospital Management System:**

2.1 Current Challenges in Hospital Management

2.1.1 Inefficient Patient Care

i. Long wait times due to manual appointment scheduling.
ii. Errors in patient records and prescriptions due to handwritten documentation.

2.1.2 Administrative & Operational Bottlenecks

i. Time-consuming paperwork increases staff workload.
ii. Difficulty in tracking medical supplies, leading to shortages or wastage. Billing and insurance claim delays due to manual processing.

2.1.3 Data Security & Accessibility Issues

i. Paper-based records are prone to loss, damage, or unauthorized access.
ii. No centralized database makes retrieving patient history slow and unreliable.
iii. Lack of HIPAA/GDPR compliance in many existing systems.

2.1.4 Lack of Real-Time Decision-Making Support

i. Hospitals struggle with delayed reporting on finances, patient flow, and treatment outcomes.

2.2 Scope

The scope of this project includes the design and implementation of a Web-based Hospital Management System that supports:

i. Patient registration and record management
ii. Appointment booking and scheduling
iii. In-patient and out-patient management
iv. Doctor and staff information management
v. Billing and payment processing
vi. Report generation and analytics

2.3 Objectives

- **Streamline Hospital Operations:**
Automate and integrate various administrative, financial, and clinical functions to improve workflow efficiency and reduce manual effort.

- **Enhance Patient Care:**
Provide healthcare professionals with quick access to accurate patient records, history, and diagnostic information to support informed and timely medical decisions.

- **Centralize Data Management:**
Maintain a secure and centralized database of all hospital-related information, including patient records, appointments, billing, inventory, and staff schedules.

- **Ensure Data Accuracy and Integrity:**
Minimize human errors by automating data entry and processing, ensuring consistent and reliable information across the system.

- **Strengthen Data Security and Privacy:**
Implement role-based access controls and data encryption to protect sensitive medical and personal information, complying with healthcare data regulations.

- **Improve Patient Experience:**
Enable features like online appointment booking, digital prescriptions, and real-time updates to make services more accessible and convenient for patients.

- **Optimize Resource Utilization:**
Track and manage hospital resources—including beds, equipment, and personnel—efficiently to reduce waste and maximize usage.

- **Enable Scalability and Integration:**
Design the system to accommodate future growth and allow integration with other systems like health insurance platforms, laboratory tools, and national health databases.

3. System Features

3.1 System Feature1

3.1.1 Description & Priority

S. No	Feature	Priority
01	Role Base Access	High
02	User Authentication	High
03	Manage Patient & Doctors	Medium
04	Make Appointments	High
05	Records Management	Medium
06	Bills& Payments	Medium
07	Inventory Management	High
08	Staff Management	Medium
09	Reports & Analysis	High

3.1.2 Stimulus/Response Sequences

Search for a particular patient history and can also search for any doctors availability, easy billing and handling of patients from any particular category.

4. Requirements:

4.1 Functional Requirements:

1) **Patient Management:**
   a) Add, update, delete, and view patient details
   b) Assign patients to wards/rooms
   c) Track patient admission and discharge

2) **Doctor Management:**
   a) Add, update, delete doctor profiles
   b) View schedules and availability
   c) Assign doctors to patients

3) **Appointment Scheduling:**
   a) Book, reschedule, and cancel appointments
   b) Notify patients and doctors of upcoming appointments
   c) Handle walk-ins and emergency cases

4) **Medical Records Management:**
   a) Store and retrieve patient medical history
   b) Upload lab reports, prescriptions, and diagnoses
   c) Ensure access control to sensitive records

5) **Billing and Payments:**
   a) Generate bills for treatments, diagnostics, and stays
   b) Apply insurance and discounts
   c) Accept multiple payment methods

6) **Inventory Management:**
   a) Track stock of medicines and medical supplies
   b) Alert for low stock levels
   c) Manage suppliers and purchase orders

7) **Staff Management:**
   a) Add and manage non-medical staff (nurses, admin, janitors)
   b) Shift scheduling and attendance tracking
   c) Payroll integration

8) **User Authentication & Role-Based Access:**
   a) Login/logout for different user types (admin, doctor, patient, staff)
   b) Access control based on roles and responsibilities

9) **Reports and Analytics:**
   a) Generate reports on revenue, patient inflow, diagnosis stats
   b) Provide dashboard views for management

4.2 Non-Functional Requirements:

1) **Performance:**
   a) Handle at least 1000 concurrent users with <2 seconds response time

2) **Scalability:**
   a) Accommodate growing numbers of patients, staff, and departments

3) **Security:**
   a) Encrypt sensitive data
   b) Regular security audits and HIPAA compliance

4) **Availability:**
   a) 99.9% uptime
   b) Backup and recovery mechanisms in place

5) **Usability:**
   a) Intuitive interface for non-technical users
   b) Support for multiple languages

6) **Maintainability:**
   a) Modular design for easy updates and bug fixes
   b) Well-documented code and workflows

7) **Compatibility:**
   a) Work across different OS and devices
   b) Integrate with other hospital systems

8) **Auditability:**
   a) Maintain logs of all system activities for auditing and traceability

5. Tools & Technologies

5.1 Frontend Tools:

- HTML (Hypertext Markup Language)
- CSS (Cascading Style Sheet)
- JAVASCRIPT

Frameworks:

- Bootstrap

5.2 Backend Tools

- PHP (Hypertext Preprocessor)
- NODEJS

Frameworks:

- Laravel

5.3 Database:

- MYSQL

5.4 IDE:

- Visual Studio Code

6. WBS Project Management

A work-breakdown structure (WBS) is a hierarchical decomposition of a project into smaller, more manageable components. It breaks down a project into smaller parts, allowing for clear definition of scope and responsibilities of each team member or task. The WBS helps to ensure that all tasks necessary for completing a project are identified, organized and assigned to the relevant team member, providing a visual representation of the project's structure and progress.

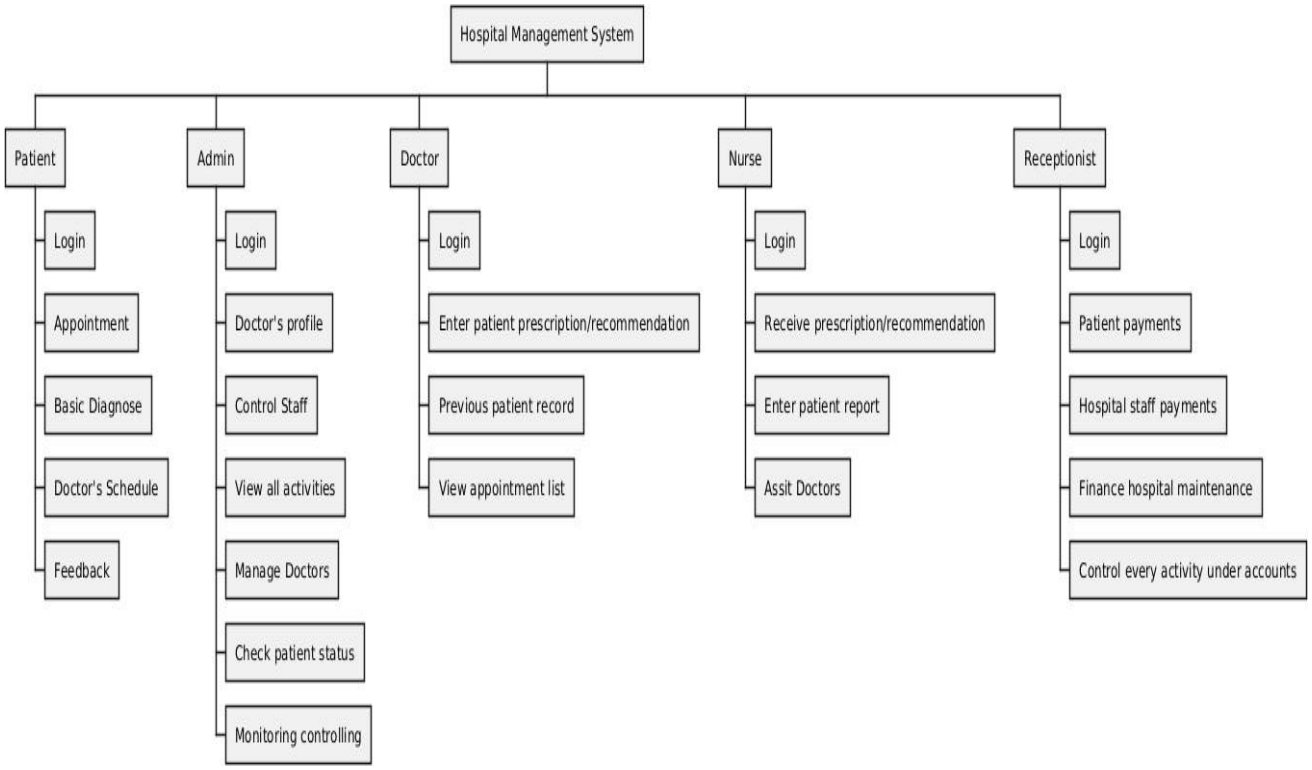


Figure 1: WBS Project Management

7. Analysis Models

7.1 Use-case Diagram:

In the Unified Modeling Language, a use case diagram can summarize the details of our system's users (also known as actors) and their interactions with the system. It is generally shown as graphic depiction of interactions among different elements in a system. Use-case diagrams specify the events in a system and how those events flow, however, use case diagrams does not describe how these events are implemented.

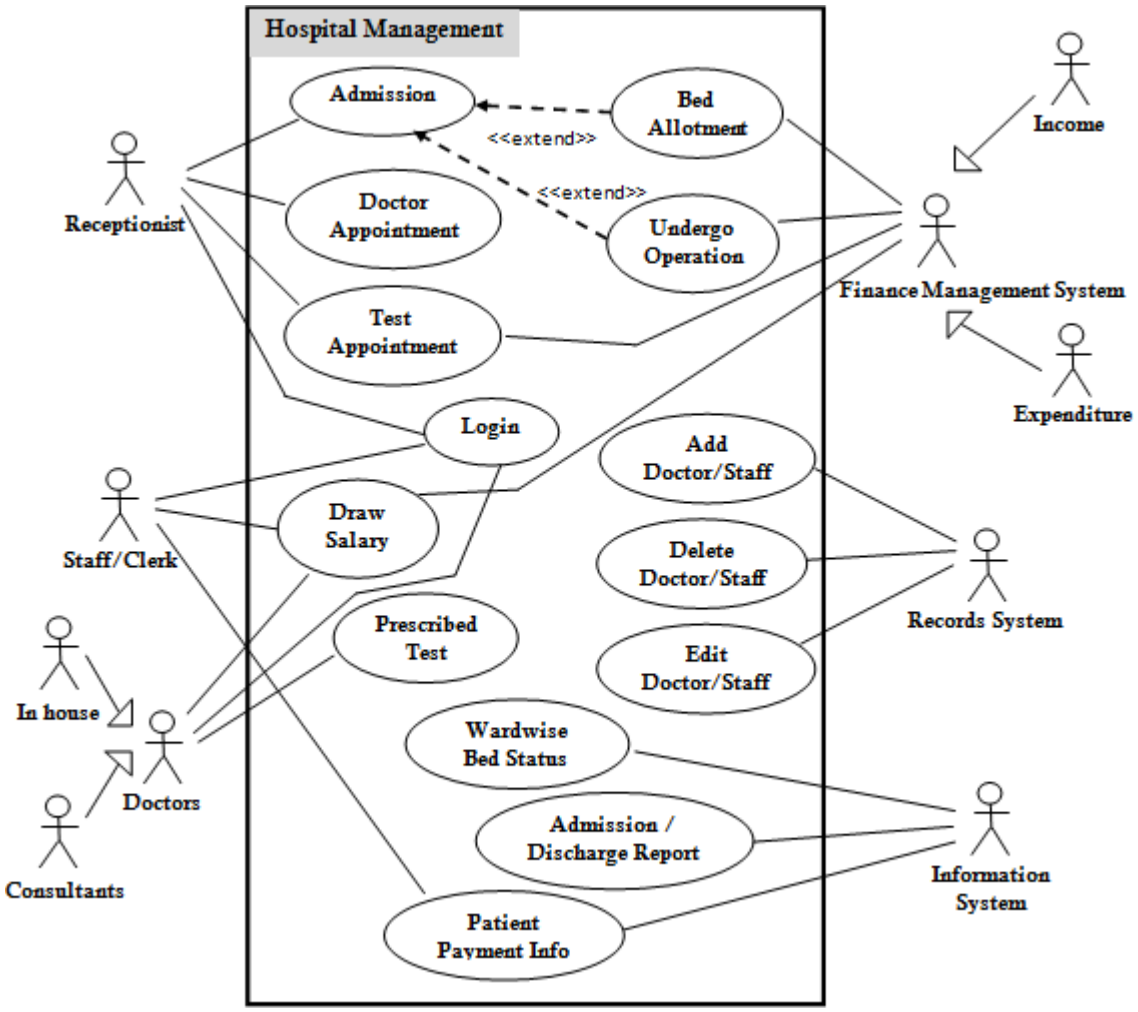


Figure 2: Use-Case Diagram

7.2 Class Diagram:

Class diagrams are blueprints of system and subsystem and illustration of relationships and source code dependencies among classes in UML. Class diagram shows a collection of classes, interfaces, associations, collaborations and constraints. It is also known as structural diagram. Class diagram consist of three sections the name of class, attributes of class and class operations. Class defines the methods and variables in an object, which is specific entity in a program or the unit of code representing that entity

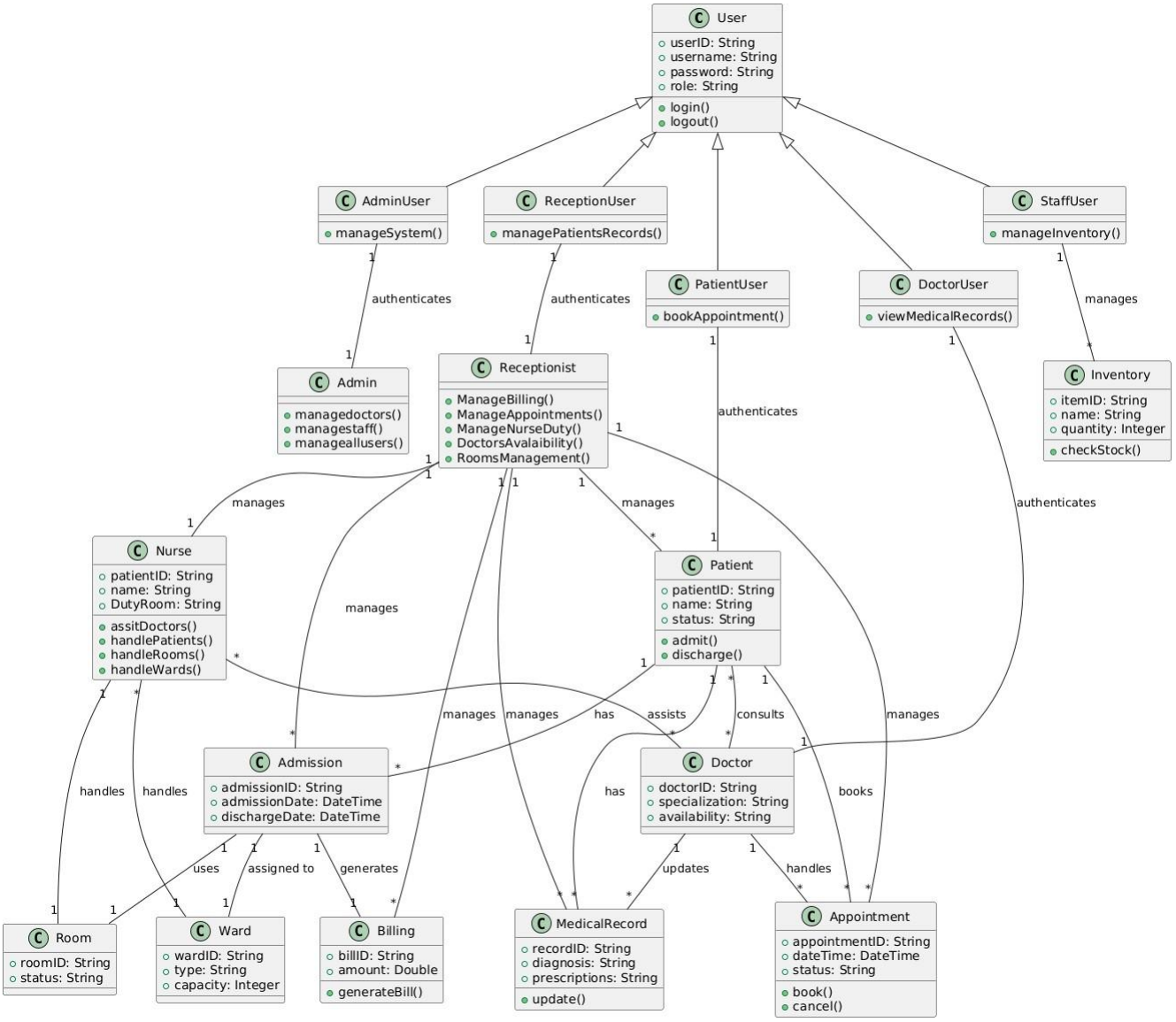


Figure 3: Class Diagram

7.3 Sequence Diagram

Sequence diagrams are an interaction diagram that illustrates the sequence of messages between object in an interaction and provide detail how operations are carried out. Sequence diagram consist of group of objects that are represented by lifeline, and the messages that they exchange over time during interaction. Sequence diagrams are time focus and they show the order of interaction visually by using the vertical axis of the diagram to represent time what messages are sent and when.

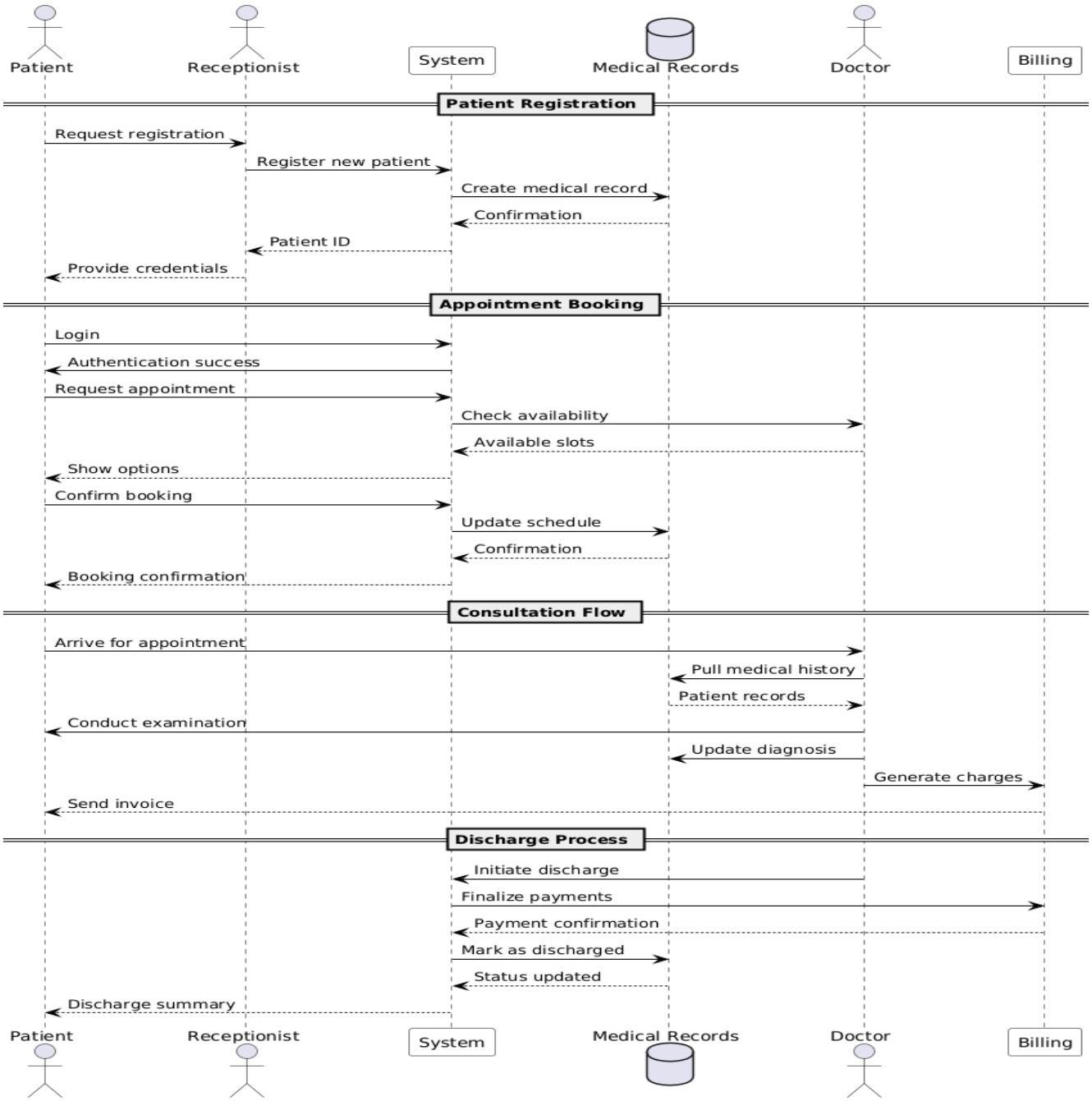


Figure 4: Sequence Diagram

7.4 Activity Diagram

An activity diagram is a type of UML flowchart that shows the flow from one activity *to another in a system or process. It's used to describe the different dynamic aspects of a system and is referred to as a behavior diagram because it describes what should happen in the modeled system.*

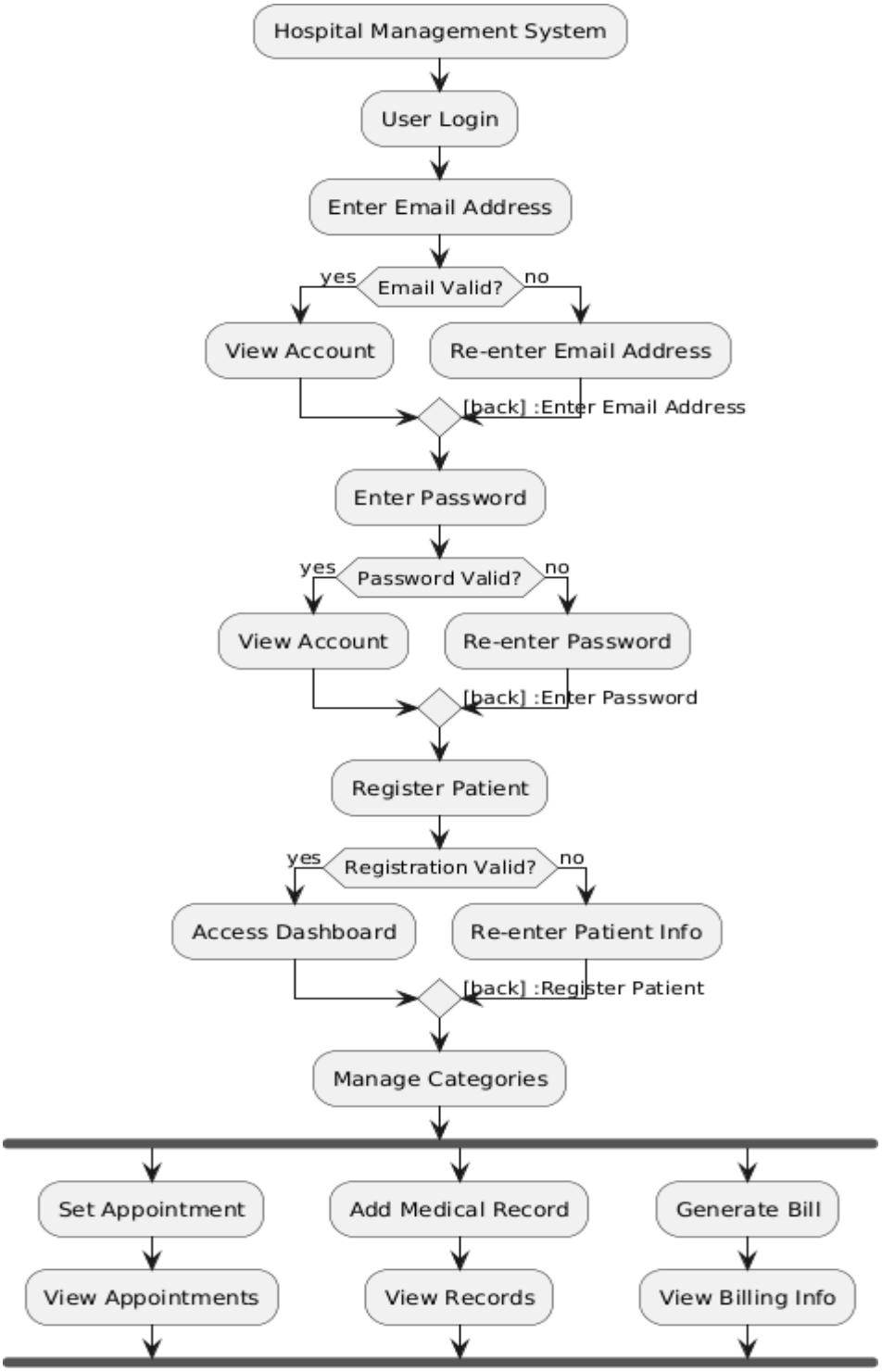


Figure 5: Activity Diagram

7.5 ER Diagram

An ERD also known as entity relationship model is a graphical representation that depicts relationship among people, objects, places, concepts or events within an information technology system. ERD's are composed of three main elements: entities, attributes and relationships.

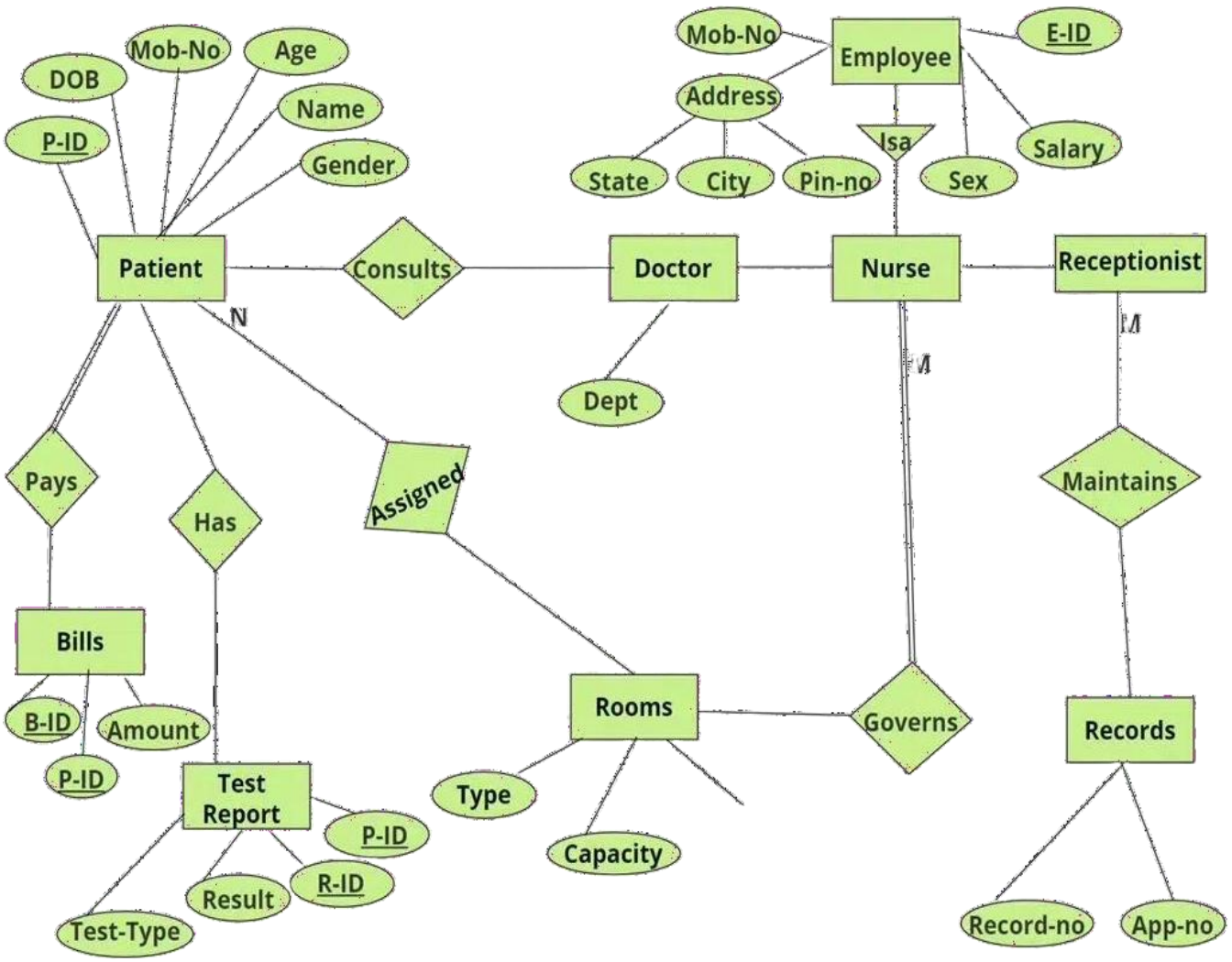
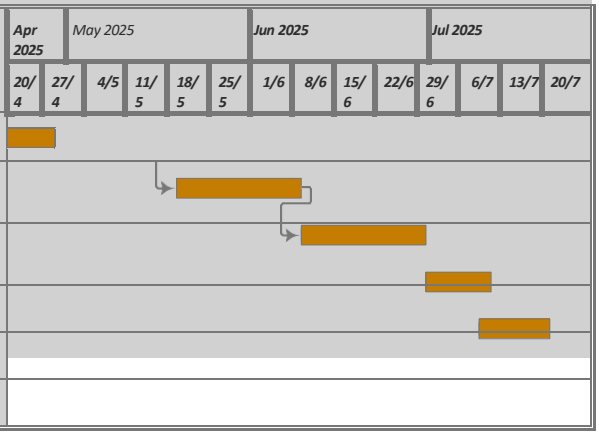


Figure 6: ER Diagram

8. Project TimeLine:

ID	Task Name	Start	Finish	Duration
1	Project Proposal	4/21/2025	4/28/2025	1.2w
2	Frontend Designing & Documentation	4/29/2025	5/19/2025	3w
3	Backend Designing and Documentation	5/20/2025	6/9/2025	3w
4	Database Implementation	6/10/2025	6/30/2025	3w
5	Testing	7/1/2025	7/11/2025	1.8w
6	Finalize Documentation	7/10/2025	7/21/2025	1.6w



9. Conclusion:

This proposal outlines the roadmap for developing a responsive, efficient, and scalable Hospital Management System. This project successfully streamlines and automates various hospital operations, including patient registration, appointment scheduling, medical records management, and billing. By integrating these functions into a centralized digital platform, the system enhances efficiency, reduces administrative workload, and improves patient care. This project demonstrates how technology can be effectively leveraged to meet the growing demands of healthcare institutions, ensuring better organization, accuracy, and accessibility of information.With a clear timeline and defined scope, this proposal serves as a roadmap to ensure the project is completed effectively and on time. Ultimately, the project contributes to enhanced patient care and a more organized healthcare environment, reflecting the critical role of technology in modern medical services.

9.1 Referenc

E<u>Websit</u>

<u>es:</u>

- ✓ https://www.w3schools.com/

Owner: Refsnes Data AS

- ✓ https://getbootstrap.com/

Developer(s): Bootstrap Core Team

- ✓ https://www.geeksforgeeks.org/

Founder: Sandeep Jain.

- ✓ https://www.scribd.com/

- ✓ https://github.com/

<u>YouTube Channels</u>

- ✓ Traversy Media
- ✓ The Net Ninja
- ✓ Web dev Simplified
- ✓ CodeWithHarry
- ✓ freeCodeCamp

Appendix A: Check List

	Check List	**Yes**	**No**
I.	Starting/Ending Dates	Yes	
II.	Project Scope	Yes	
III.	Product modules (covering all aspects of scope)	Yes	
IV.	System Features (covering scope)	Yes	
V.	Interface Requirements	Yes	
VI.	Non-Functional Requirements	Yes	
VII.	WBS	Yes	
VIII.	Tools and Technologies Detail (for implementation)	Yes	
IX.	Plagiarism Report		No

Appendix B: Supervisory Committee

For Approval of any two Consultant Teachers	
Teacher Consulted **Name: _____.** **Designation: _____.** **Comments: _____** **_____** **_____** **_____** **_____** **Signature: _____**	**Teacher Consulted** **Name: _____.** **Designation: _____.** **Comments: _____** **_____** **_____** **_____** **_____** **Signature: _____**

(For office use only)

Date:_____

- **Approved: _____**
- **Group ID: _____**
- **☐ Meeting Required: Date:_____Time:_____Place: _____**
- **Rejected:**

Remarks:

_____
_____
_____
_____

Project Title (if Revised):

_____
_____
_____
_____

Project Coordinator: _____